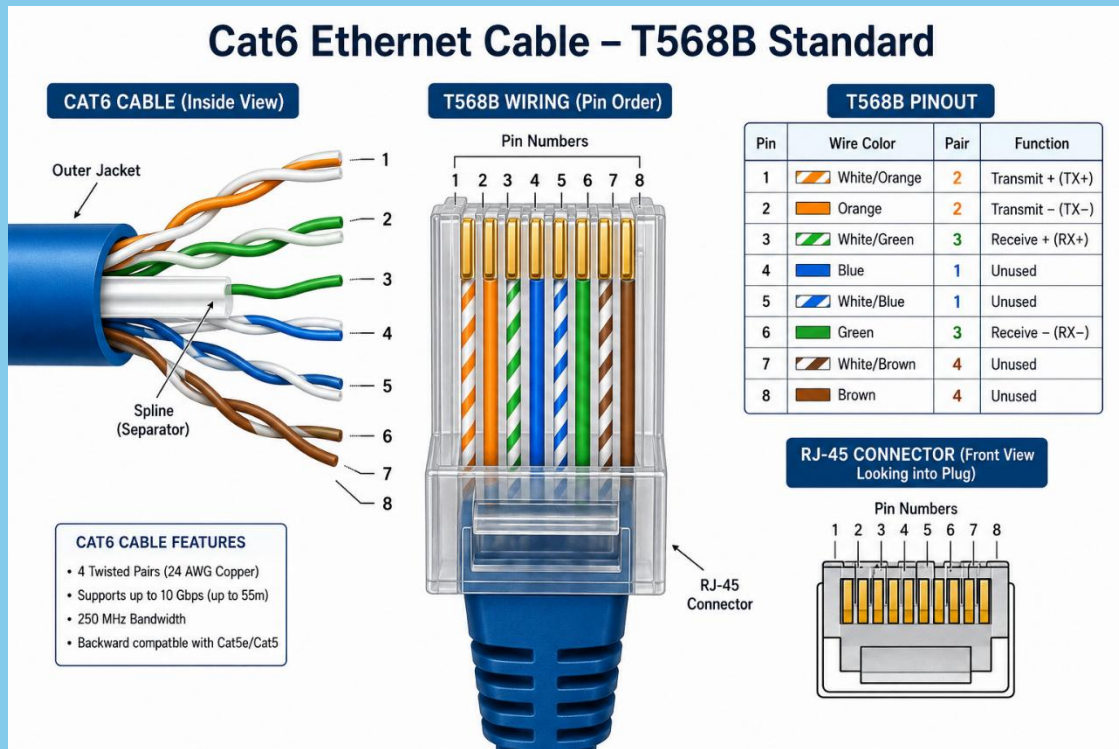


Network Cable and Connectors

1. Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.



2. List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why..

Network Cable	Real-World Application / Device	Why it is used
Cat5e	Wi-Fi Router	Provides reliable Ethernet connectivity for home/office networks and supports speeds up to 1 Gbps.
Cat6	Zomato POS Terminal	Provides faster and more reliable wired connectivity, suitable for transferring billing and order data quickly.
Fiber Optic	IPL Streaming Server	Supports very high bandwidth and long-distance communication, making it suitable for delivering high-quality live video to many users.

3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Step 1 – Measure and Cut the Cable

First, I measured the required length of the Ethernet cable. After measuring, I used the cutter on the crimping tool to cut the cable neatly.

A clean cut makes the next steps much easier.

Step 2 – Strip the Outer Jacket

Next, I used the wire stripper to remove about **2 to 3 cm (1 inch)** of the outer plastic jacket.

While removing the jacket, I made sure not to damage the insulation of the eight small wires inside.

If any wire gets cut or scratched, the cable should be cut again and stripped properly.

Step 3 – Separate the Twisted Pairs

Inside the cable there were four twisted pairs.

- Orange pair
- Green pair
- Blue pair
- Brown pair

I carefully untwisted each pair only as much as needed because keeping the twists close to the connector helps reduce signal interference.

Step 4 – Arrange the Wires

I straightened all eight wires using my fingers and arranged them in the **T568B standard**.

The order was:

White/Orange

Orange

White/Green

Blue

White/Blue

Green

White/Brown

Brown

I checked the color order twice before moving to the next step because even one wire in the wrong position will make the cable fail.

Step 5 – Trim the Wires

After arranging the wires, I held them tightly together and cut all eight wires to the same length.

The cut had to be perfectly straight so that every wire would reach the end of the RJ-45 connector equally. Uneven wire lengths are a common cause of failed crimps.

Step 6 – Insert the Wires into the RJ-45 Connector

I held the RJ-45 connector with the clip facing downward.

Then I slowly pushed all eight wires into the connector while making sure:

- Every wire entered its correct channel.
- Every wire reached the front end of the connector.
- The outer cable jacket also entered slightly inside the connector so it could be secured by the strain relief.

I looked through the transparent connector to confirm the wire colors were still in the correct order.

Step 7 – Crimp the Connector

After confirming everything was correct, I inserted the RJ-45 connector into the crimping tool.

I squeezed the handles firmly until the ratchet released.

This pushed the gold pins into the wires and locked the connector onto the cable.

4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Connection	Cable	Connector	Reason
Internet/Network	Cat6 Ethernet cable	RJ-45	Provides fast, stable, low-latency network connectivity for gaming consoles and streaming devices.
Screen to Console	HDMI 2.1 cable	HDMI connector	Supports high-resolution video and high refresh rates with low delay, suitable for modern gaming consoles and displays.
Long-distance screen connection	Fiber-optic HDMI cable	HDMI connector	Useful when screens are far from the gaming/streaming setup because it can carry high-bandwidth video over longer distances.

5. Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.



Type of cable: Cat6 Ethernet Cable

Connector type: RJ-45 (Registered Jack-45)

Where it is used: Connected to the **LAN port of a WiFi router.**

Devices it connects:

- Wi-Fi Router ↔ Desktop Computer
- Wi-Fi Router ↔ Laptop
- Wi-Fi Router ↔ Network Switch
- Wi-Fi Router ↔ Smart TV or Gaming Console
- Provides a **fast and stable wired network connection.**
- Supports speeds up to **1 Gbps** (and up to **10 Gbps** over shorter distances with Cat6).
- Offers lower latency and greater reliability than Wi-Fi, making it ideal for gaming, video streaming, and file transfers.